

DEL BONITA AGDM, AB, Canada

WMO: 712240

Lat: 49.0500N

Lon: 112.8099W

Elev: 4298

StdP: 12.55

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.6	-11.0	-22.8	1.8	-15.9	-17.0	2.5	-10.7	37.7	39.6	32.8	34.2	8.7	320	0.773

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.0	86.1	58.1	82.7	57.4	79.2	56.7	61.9	77.3	60.1	76.0	58.5	74.9	10.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.9	81.2	66.5	54.4	73.9	64.4	52.2	68.2	62.8	29.9	77.3	28.5	75.8	27.3	75.0	68.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.5	28.2	24.8	DB	-23.7	91.7	7.3	3.6	-29.0	94.2	-33.3	96.3	-37.4	98.3	-42.7	100.9	(4)
(5)			WB	-24.0	66.0	7.3	1.9	-29.3	67.3	-33.5	68.4	-37.6	69.5	-42.9	70.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.2	22.7	20.6	29.7	38.3	47.6	54.8	62.7	61.4	52.5	40.9	29.5	20.7
	DBStd	18.28	13.89	14.32	11.96	8.95	8.03	6.09	5.93	6.63	8.88	10.27	12.87	13.98
	HDD50	4736	848	824	628	360	143	20	1	5	74	308	618	907
	HDD65	9128	1313	1244	1093	801	540	310	115	147	382	746	1065	1372
	CDD50	1174	0	0	1	9	69	165	393	358	149	27	3	0
	CDD65	91	0	0	0	0	1	5	42	36	7	0	0	0
	CDH74	2126	0	0	0	2	35	117	923	807	235	7	0	0
	CDH80	610	0	0	0	0	2	20	299	237	52	0	0	0
	WSAvg	11.9	14.1	12.2	12.9	12.5	11.7	11.1	9.5	9.5	10.7	12.3	13.6	13.4
	PrecAvg	14.9	0.5	0.4	0.7	1.2	2.2	3.5	1.6	1.4	1.3	0.9	0.6	0.6
(15) Precipitation	PrecMax	22.8	0.8	0.9	1.6	2.4	4.2	8.9	5.5	3.3	3.7	2.7	1.3	1.1
	PrecMin	8.0	0.1	0.0	0.2	0.5	0.6	0.6	0.1	0.0	0.1	0.2	0.2	0.0
	PrecStd	3.5	0.2	0.2	0.4	0.5	1.0	2.0	1.2	0.8	0.9	0.6	0.3	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.4	52.2	59.6	70.6	78.2	83.1	91.5	89.9	85.3	72.6	62.4	49.5
		MCWB	39.9	38.6	43.4	48.8	55.3	61.3	60.5	57.9	55.2	50.6	45.5	38.8
	2%	DB	45.8	45.6	53.8	63.6	73.0	77.1	86.9	85.8	80.9	66.6	54.3	44.1
		MCWB	37.6	36.6	41.3	45.9	52.3	57.8	59.1	57.2	54.4	48.7	42.9	36.5
	5%	DB	42.2	41.9	49.3	58.8	68.6	73.4	83.2	82.4	75.7	61.1	49.5	40.7
		MCWB	35.4	34.5	39.2	43.5	50.4	56.0	58.9	56.5	53.1	46.9	39.8	34.8
	10%	DB	39.4	38.5	45.3	53.8	63.9	69.4	79.2	78.6	70.1	55.9	45.3	38.0
		MCWB	34.0	32.6	37.0	41.3	48.5	54.2	58.3	55.8	51.1	44.0	37.5	32.9
	0.4%	WB	41.6	40.3	44.8	50.0	56.1	63.6	65.7	61.9	58.4	53.2	47.6	40.7
		MCDB	49.0	49.1	57.1	65.3	73.7	77.5	78.2	78.4	74.8	68.2	58.7	47.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.8	37.3	42.2	46.8	53.3	60.2	62.9	59.9	55.9	49.8	43.4	37.1
		MCDB	44.5	44.7	51.8	61.8	69.9	73.8	77.7	76.8	74.8	64.4	53.6	42.7
	5%	WB	35.8	34.9	39.9	44.4	51.3	57.6	61.1	58.5	54.0	47.2	40.4	35.3
		MCDB	41.5	41.5	48.5	57.4	66.1	70.1	76.8	76.4	73.2	59.6	48.6	40.5
	10%	WB	34.1	32.7	37.2	41.8	49.4	55.3	59.6	57.1	51.9	44.8	37.8	33.0
		MCDB	39.4	38.1	44.6	52.8	62.1	67.0	75.9	75.1	69.0	54.7	44.7	38.0
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	18.7	18.6	20.8	21.7	22.1	28.0	27.7	24.3	21.2	18.0	17.0
		MCDBR	17.9	20.8	23.5	29.1	29.9	28.8	33.8	34.5	34.1	27.4	21.4	17.6
		MCWBR	12.1	13.7	13.7	14.8	13.7	13.8	13.6	13.0	14.0	13.7	12.6	12.3
	5% WB	MCDBR	17.7	21.3	22.4	27.3	26.7	26.1	29.5	29.6	30.3	26.2	20.6	17.5
		MCWBR	12.7	15.0	13.8	14.5	12.7	13.6	13.7	12.3	13.1	13.7	12.7	12.8
(40) Clear-Sky Solar Irradiance	taub		0.237	0.246	0.272	0.306	0.317	0.315	0.316	0.334	0.300	0.269	0.247	0.224
	taud		2.444	2.469	2.486	2.416	2.443	2.496	2.499	2.420	2.491	2.560	2.513	2.452
	Ebn at Noon		272	296	303	300	299	299	297	287	288	283	267	265
	Edh at Noon		19	24	28	34	34	33	32	33	28	22	18	16
(44) All-Sky Solar Radiation	RadAvg		393	710	1096	1510	1799	2013	2258	1812	1288	791	437	315
	RadStd		32	53	99	98	160	159	130	117	105	76	33	28

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page